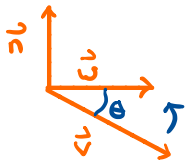
13.4 The cross product

If $\vec{v}$ and $\vec{w}$ are in $\mathbb{R}^3$, their cross product is defined

$$\vec{v} \times \vec{w} = \|\vec{v}\| \, \|\vec{w}\| \sin(\theta) \, \vec{n}$$ ← "geometric definition"

where θ is the angle between $\vec{v}$ and $\vec{w}$ and $\vec{n}$ is a unit vector orthogonal to both $\vec{v}$ and $\vec{w}$ determined by the right-hand rule.



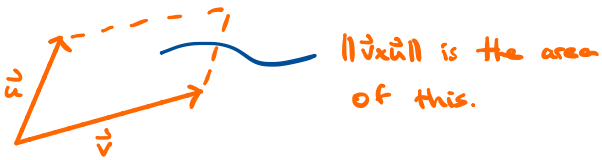
Equivalently, if $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$ and $\vec{w} = w_1\vec{i} + w_2\vec{j} + w_3\vec{k}$, then

$$\vec{v} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (v_2w_3 - w_2v_3)\vec{i} - (v_1w_3 - w_1v_3)\vec{j} + (v_1w_2 - w_1v_2)\vec{k}$$

↖ "algebraic definition"

→ Do problems 1, 2.

<u>Fact</u> The magnitude $\|\vec{v} \times \vec{u}\|$ is the area of the parallelogram "spanned" by $\vec{v}$ and $\vec{u}$:



→ Do problems 3-5.